Neural Operator

29.08.2026

Iterative Sequence von Funktionen: $v_0 \mapsto v_1 \mapsto \dots \mapsto v_{T-1} \mapsto v_T$ "Layer"

$P(a(x)) = v_0$ lokale Transformationen $Q(v_T(x)) = u(x)$

Komposition von...

$$v_{t+1}(x) = \sigma\left(W v_t(x) + (\mathcal{K}(a;\phi) v_t)(x)\right)$$

- σ: lokale, nicht-lineare Aktivierung (indiv. pro v_j)
- W: lineare Transformation
- $\mathcal{K}(a;\phi)$: globale Integraloperator (für alle v_j)

$\mathcal{K}: A \times \Theta_{\mathcal{K}} \to \mathcal{L}(\mathcal{U}(D;\mathbb{R}^{d_v}), \mathcal{U}(D;\mathbb{R}^{d_v}))$ (linear Operator)

$$\int_D \kappa(x, y, a(x), a(y); \phi) v_t(y) dy =: (\mathcal{K}(a;\phi) v_t)(x) \quad (\forall x \in D, y \in D \text{ kompakt})$$

Kernel κ parametrisiert via Neuronales Netz, d.h. mit festen Parametern $\phi \in \Theta_{\mathcal{K}}$

Fourier Neural Operator: $(\mathcal{K}(a;\phi) v_t)(x) = \mathcal{F}^{-1}(F(\kappa_\phi) \cdot \mathcal{F}(v_t))(x)$

Fourier transform of func. f for $j = 1, \dots, d_v$:

$$(\mathcal{F} f)_j(k) = \int_D f_j(x) e^{-2i\pi\langle x,k\rangle} dx$$

...and it's inverse: $(\mathcal{F}^{-1} f)_j(x) = \int_D f_j(k) e^{2i\pi\langle x,k\rangle} dk$

Frequency Mode $k \in D$

$\Rightarrow$ Fourier Integral operator $\mathcal{K}$: $(\mathcal{K}(\phi) v_t)(x) = \mathcal{F}^{-1}(R_\phi \cdot (\mathcal{F} v_t))(x)$

Fourier Transform of a periodic function: $\kappa: \bar{D} \to \mathbb{R}^{d_v \times d_v}$ parametrized by $\phi \in \Theta_{\mathcal{K}}$

(max. nr. of freq. modes)

$R_\phi \in \mathbb{C}^{k_{max} \times d_v \times d_v}$

Diskretisierung von k: $k \in \mathbb{Z}^d$ or $k = 1, \dots, k_{max}$

$\Rightarrow$ Faltung $(R \cdot (\mathcal{F} v_t)_{k,l}$ becomes discrete)

$$= \sum_{j=1}^{d_v} R_{k,j,l} (\mathcal{F} v_t)_{k,j}, \quad j = 1, \dots, d_v$$

Diskretisierung
- uniform ← required for efficient computation $\Rightarrow$ FFT ($\hat{\mathcal{F}}$) (majority computation lies on computing $\mathcal{F}(v_t)$ and inverse)
- Resolution $n = s_1 \times \dots \times s_d$

direct parametrization of R (best)
- -||- similar
- -||- worse

$$(\hat{\mathcal{F}} f)_l(k) = \sum_{x_1=0}^{s_1-1} \cdots \sum_{x_d=0}^{s_d-1} f_l(x_1, \dots, x_d) e^{-2\pi i \langle x, k\rangle}$$

$$(\hat{\mathcal{F}}^{-1} f)_l(x) = \sum_{k_1=0}^{s_1-1} \cdots \sum_{k_d=0}^{s_d-1} f_l(k_1, \dots, k_d) e^{2\pi i \langle x, k\rangle}$$

truncated modes of FFT: with $\langle x, k\rangle := \sum_{j=1}^{d} \frac{x_j k_j}{s_j}$ for $l = 1, \dots, d$

$Z_{k_{max}} = \{(k_1, \dots, k_d) \in \mathbb{Z}_{s_1} \times \dots \times \mathbb{Z}_{s_d} \mid k_j \le k_{max,j} \text{ or } s_j - k_j \le k_{max,j} \text{ for } j = 1, \dots, d\}$

$\Rightarrow R \in (s_1 \times \dots \times s_d \times d_v \times d_v)$-Tensor; Zahl unserer Parameter

In practice but: